NORTH BENGAL INTERNATIONAL UNIVERSITY

Lab Report : 3

TOPIC : WRITE A PROGRAM FOR USER-DEFINED FUNCTION TYPES AND FUNCTION WITH A PASS-BY REFERENCE IN C++

COURSE CODE : CSE 0612-2122

COURSE TITLE : OBJECT ORIENTED PROGRAMMING SESSIONAL.

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• WRITE A PROGRAM FOR USER-DEFINED FUNCTION TYPES AND FUNCTION WITH A PASS-BY REFERENCE IN C++:

program to passed argument with a return value in C++ :

Code:

```
#include <iostream>

int square(int num) {
    return num * num;
}

int main() {
    int number;

    std::cout << "Enter a number: ";
    std::cin >> number;

    int result = square(number);

    std::cout << "The square of " << number << " is " << result <<
std::endl;

    return 0;
}
```

Output:

```
OUTPUT    TERMINAL

Enter a number: 5
The square of 5 is 25
PS C:\Users\Lenovo\Documents\c++>
```

program to passed no argument with a no return value in C++ :

Code:

```
#include <iostream>

void greet() {
    std::cout << "Hello! Welcome to the C++ world!" << std::endl;
}

int main() {

    greet();

    return 0;
}
```

Output:

```
OUTPUT    TERMINAL

Hello! Welcome to the C++ world!
PS C:\Users\Lenovo\Documents\c++>
```

program to passed no argument but with a return value in C++:

Code:

```
#include <iostream>
#include <ctime>

int getCurrentYear() {
    time_t now = time(0);
    tm *ltm = localtime(&now);
    return 1900 + ltm->tm_year;
}

int main() {

    int currentYear = getCurrentYear();

    std::cout << "The current year is: " << currentYear <<
std::endl;

    return 0;
}
```

Output:

```
OUTPUT    TERMINAL

The current year is: 2024
PS C:\Users\Lenovo\Documents\c++>
```

program to passed argument but no return value in C++:

Code:

```
#include <iostream>

// Function that takes two integers as arguments and prints their
sum
void printSum(int a, int b) {
    int sum = a + b;
    std::cout << "The sum of " << a << " and " << b << " is " << sum
<< std::endl;
}

int main() {
    int num1, num2;
```

```
    std::cout << "Enter the first number: ";
    std::cin >> num1;

    std::cout << "Enter the second number: ";
    std::cin >> num2;

    printSum(num1, num2);

    return 0;
}
```

Output:

```
OUTPUT    TERMINAL

Enter the first number: 5
Enter the second number: 5
The sum of 5 and 5 is 10
PS C:\Users\Lenovo\Documents\c++>
```

Program to pass by Reference:

Code:

```
#include <iostream>

void swap(int &a, int &b) {
    int temp = a;
    a = b;
    b = temp;
}
int main() {
    int num1, num2;

    std::cout << "Enter the first number: ";
    std::cin >> num1;

    std::cout << "Enter the second number: ";
    std::cin >> num2;

    std::cout << "Before swapping: num1 = " << num1 << ", num2 = "
<< num2 << std::endl;

    swap(num1, num2);

    std::cout << "After swapping: num1 = " << num1 << ", num2 = " <<
num2 << std::endl;

    return 0;
}
```

Output:

```
OUTPUT    TERMINAL

Enter the first number: 5
Enter the second number: 7
Before swapping: num1 = 5, num2 = 7
After swapping: num1 = 7, num2 = 5
PS C:\Users\Lenovo\Documents\c++>
```